Reproducible Experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One readable experiment-identity and replay record.

Trace output versions,audit summary statistics and name missing reproduction evidence.

Keep one habit: W1–W3/W5 → selected seed summary → project run identity → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

Metrics Match Consequences

> **My remembered idea, example and correction**

Hold Out Test Data

> **My remembered idea, example and correction**

Mean, Median, and Baselines

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main declared seed labels 11/22/33 have14/16/12 correct of20, mean 70%, range 60–80%or20 percentage points. These classifier outcomes are supplied, not a rerunnable training algorithm. Synthetic V01–V20 prediction lists reproduce their arithmetic only. Separate pictureseed 4/5/6 has6/8/7 correct of10, also mean 70%; deleting 60%leaves mean 75%but incomplete reporting.

  Complete deterministic policy code inputsA4/1, B2/4; v1 weights2/1 outputs9/8, v2 weights1/2 outputs6/10. No random operation or accuracy claim in that program.

- Reference for this exercise: open this week's student lesson, "Reproducible Experiments", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Calculate the worked mean and range in percentage points.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main declared seed labels 11/22/33 have14/16/12 correct of20, mean 70%, range 60–80%or20 percentage points. These classifier outcomes are supplied, not a rerunnable training algorithm. Synthetic V01–V20 prediction lists reproduce their arithmetic only. Separate pictureseed 4/5/6 has6/8/7 correct of10, also mean 70%; deleting 60%leaves mean 75%but incomplete reporting.

  Complete deterministic policy code inputsA4/1, B2/4; v1 weights2/1 outputs9/8, v2 weights1/2 outputs6/10. No random operation or accuracy claim in that program.

- A file contains only **seed** 22 and 80 percent. Name four missing items.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main declared seed labels 11/22/33 have14/16/12 correct of20, mean 70%, range 60–80%or20 percentage points. These classifier outcomes are supplied, not a rerunnable training algorithm. Synthetic V01–V20 prediction lists reproduce their arithmetic only. Separate pictureseed 4/5/6 has6/8/7 correct of10, also mean 70%; deleting 60%leaves mean 75%but incomplete reporting.

  Complete deterministic policy code inputsA4/1, B2/4; v1 weights2/1 outputs9/8, v2 weights1/2 outputs6/10. No random operation or accuracy claim in that program.

- Does identical **output** prove a valid experiment?

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main declared seed labels 11/22/33 have14/16/12 correct of20, mean 70%, range 60–80%or20 percentage points. These classifier outcomes are supplied, not a rerunnable training algorithm. Synthetic V01–V20 prediction lists reproduce their arithmetic only. Separate pictureseed 4/5/6 has6/8/7 correct of10, also mean 70%; deleting 60%leaves mean 75%but incomplete reporting.

  Complete deterministic policy code inputsA4/1, B2/4; v1 weights2/1 outputs9/8, v2 weights1/2 outputs6/10. No random operation or accuracy claim in that program.

- One planned **seed** fails before **prediction**. How should it appear in the record?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use all three declared runs with the same 12-case denominator. Report mean run accuracy and the max-minus-min range in percentage points. The task supplies aggregate outcomes only; it does not supply per-case predictions or an executable classifier, so do not claim that the arithmetic reconstructs training. Retain every run and state the limited evidence.

- Declared **seed**s 1, 2, 3 produce 9/12, 8/12, 10/12 correct. Compute the mean accuracy and range.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply independently

> **My independent explanation, trace and limits**
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MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Give every Plant Classifier experiment an ID and immutable record linking dataset, split, code, settings, **prediction**s, and metrics.

- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 22

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Improve a record for Experimental records and **seed**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for Experimental records and **seed**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Two teams use the same **random seed** but different label files. Their results differ. List the records needed to reproduce the experiment.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A report says accuracy85% without data split, code **version**, **seed** or pre**process**ing.
- Specify **reproducibility** records.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Two runs use different random splits and different feature sets. Their scores differ. What must be recorded, and why is the cause unclear?
- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 26 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Two teams use the same **random seed** but different label files. Their results differ. List the records needed to reproduce the experiment.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Declared **seed**s 1, 2, 3 produce 9/12, 8/12, 10/12 correct. Compute the mean accuracy and range.

My reasoning and evidence

▼ **Extension · Versioned experiments — W1 Record a run**

REFERENCE · SECTION 17

Extension · Versioned experiments — W1 Record a run

> Retained extension task with its own values and teacher key.

Extension question and evidence

- List five items needed to reproduce a scoring run.

My reasoning and evidence

▼ **Extension · Versioned experiments — W2 Compute a comparison**

REFERENCE · SECTION 18

Extension · Versioned experiments — W2 Compute a comparison

> Retained extension task with its own values and teacher key.

Extension question and evidence

- A has fun 3 and learning 2. Score it under weights (2,1) and (1,2).

My reasoning and evidence

▼ Extension · Versioned experiments — W3 Repair an experiment

REFERENCE · SECTION 19

Extension · Versioned experiments — W3 Repair an experiment

> Retained extension task with its own values and teacher key.

Extension question and evidence

- A learner changes weights, **input** ratings and the time limit together. Why is attribution difficult?

My reasoning and evidence

▼ Extension · Versioned experiments — W4 Design independently

REFERENCE · SECTION 20

Extension · Versioned experiments — W4 Design independently

> Retained extension task with its own values and teacher key.

Extension question and evidence

- Plan two **version**s for C (fun 5, learning 1) and D (fun 2, learning 5), using weights (2,1) and (1,2). Record expected scores, ranks, fixed data and a separate final-case plan.

My reasoning and evidence

▼ **Extension · Versioned experiments — W5 Protect evaluation**

REFERENCE · SECTION 21

Extension · Versioned experiments — W5 Protect evaluation

> Retained extension task with its own values and teacher key.

Extension question and evidence

- You revise the policy after reading final-case failures. Are those cases still untouched **holdout evidence**?

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 23

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Record variation across declared **seed**s instead of selecting the prettiest run.
- Same fixed validation set of10 cases; declared runs **seed** 4=6 correct, **seed** 5=8, **seed** 6=7. All three are reported.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- What should accompany the mean **70%**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/26

Reproducible Experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Keep run identities

Record seeds4, 5, 6 with 6/10, 8/10, 7/10.

2. Summarise variation

Mean accuracy is 70%; range is 60% to80%.

3. Preserve context

Record data version, split, code, settings and seeds, not just the winning80%.

CHANGE ONE THING

The report deletes the 60% run.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 24

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 25

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Record variation across declared **seed**s instead of selecting the prettiest run.
- Which labelled step matches this detail? Record **data version**, split, code, settings and **seed**s, not just the winning**80%**.
- 1. Preserve context
- 2. Keep run identities
- 3. Summarise variation
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Preserve context Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. Read the picture in order.

Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The report deletes the **60%** run. Which conclusion is supported by the supplied facts?
- 1. The remaining mean rises to75%, but the reported comparison becomes selectively incomplete.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> _____
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> _____
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> _____
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> _____

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The remaining mean rises to75%, but the reported comparison becomes selectively incomplete.

Removing an unfavourable declared run changes the **evidence** rather than improving the method.

Repair and keep the record

Review Metrics Match Consequences (ai-4/25). Goal: Apply a declared score threshold including equality. Short practice: Which labels determine the false-alarm count at threshold 0.4?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Hold Out Test Data (ai-4/23). Goal: Freeze the full evaluation procedure before testing. Short practice: What is needed for an honest final claim about the revised **version**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Mean, Median, and Baselines (ai-4/20). Goal: Fit mean and median constant baselines on **training** targets. Short practice: Does either original baseline win under evaluation MAE?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

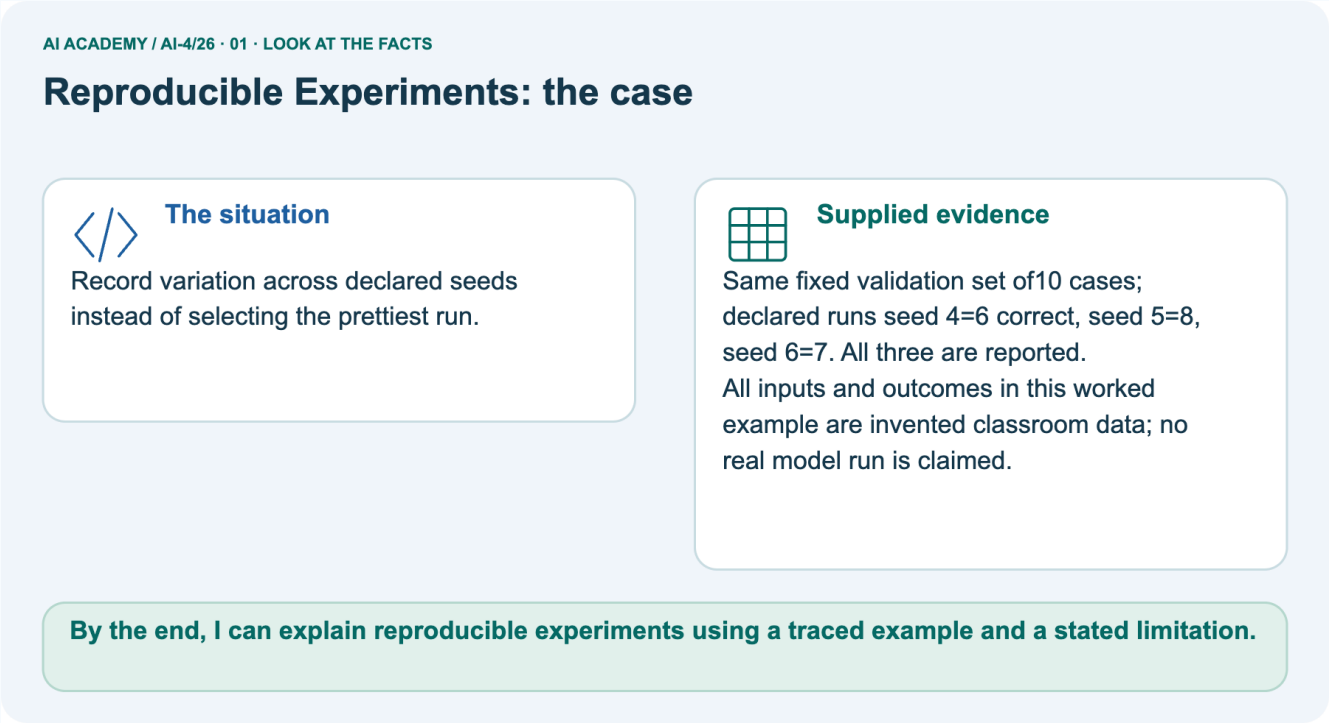
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 26

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Reproducible Experiments: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

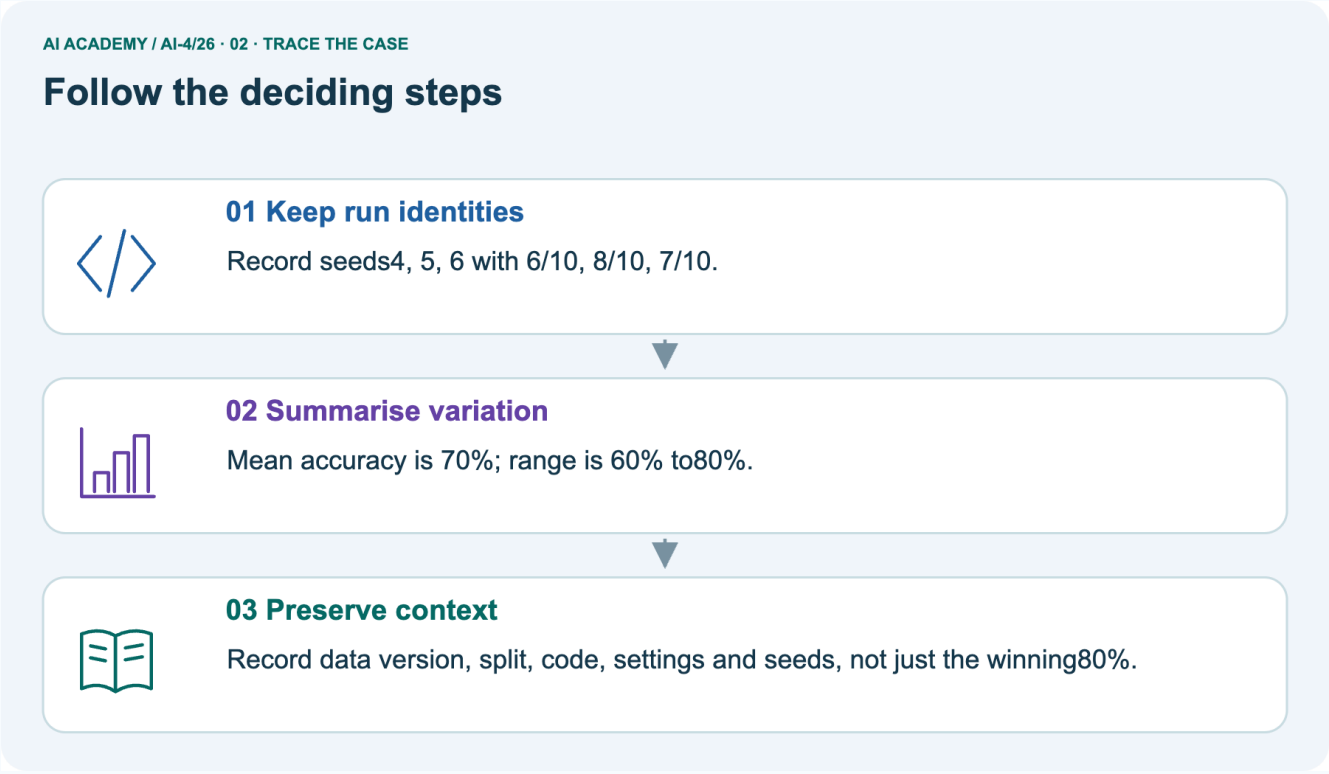
Record variation across declared **seed**s instead of selecting the prettiest run. Supplied facts: Same fixed validation set of10 cases; declared runs **seed** 4=6 correct, **seed** 5=8, **seed** 6=7. All three are reported.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep run identities: Record **seed**s4, 5, 6 with 6/10, 8/10, 7/10. → Summarise variation: Mean accuracy is **70%**; range is **60%** to**80%**. → Preserve context: Record **data version**, split, code, settings and **seed**s, not just the winning**80%**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. All declared seeds



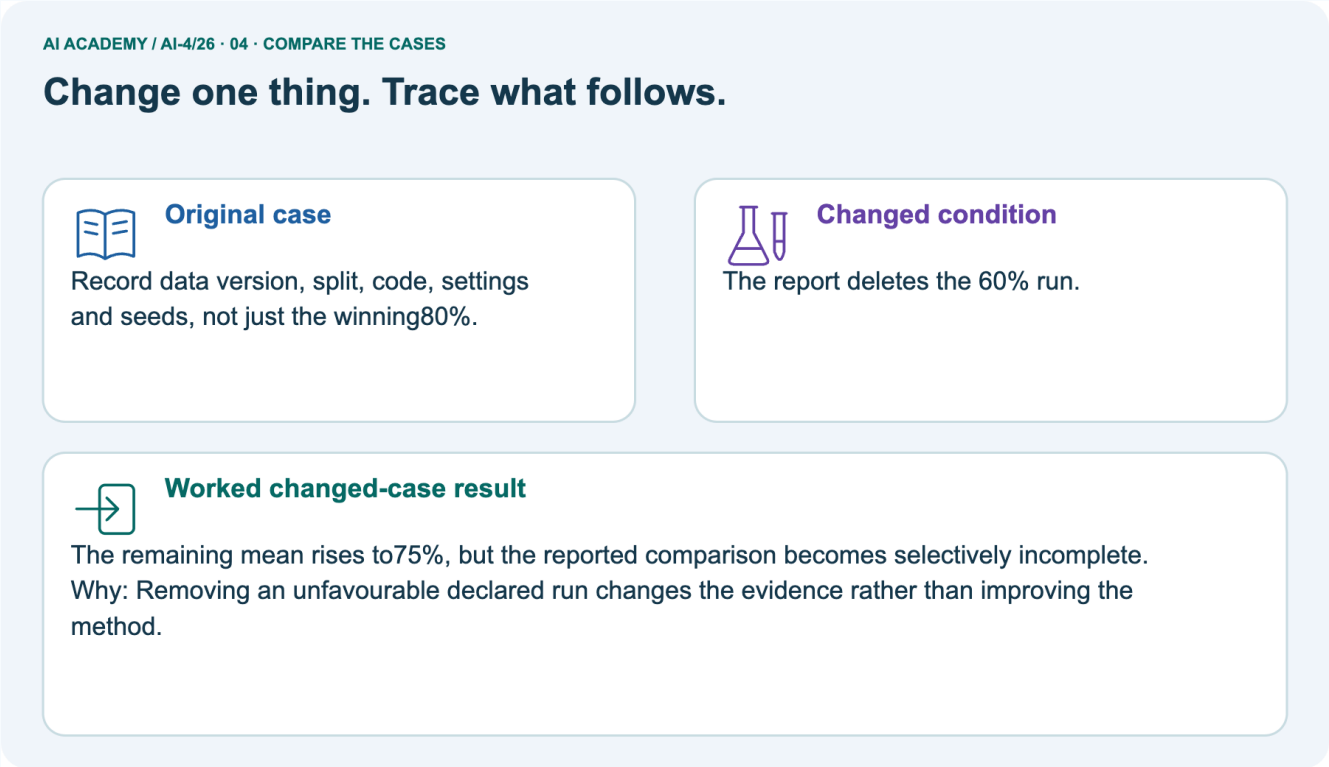
Follow the labelled picture with the matching explanation. Open the vector diagram

All declared **seed**s. Units: accuracy percent. **seed** 4: 60. **seed** 5: 80. **seed** 6: 70.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

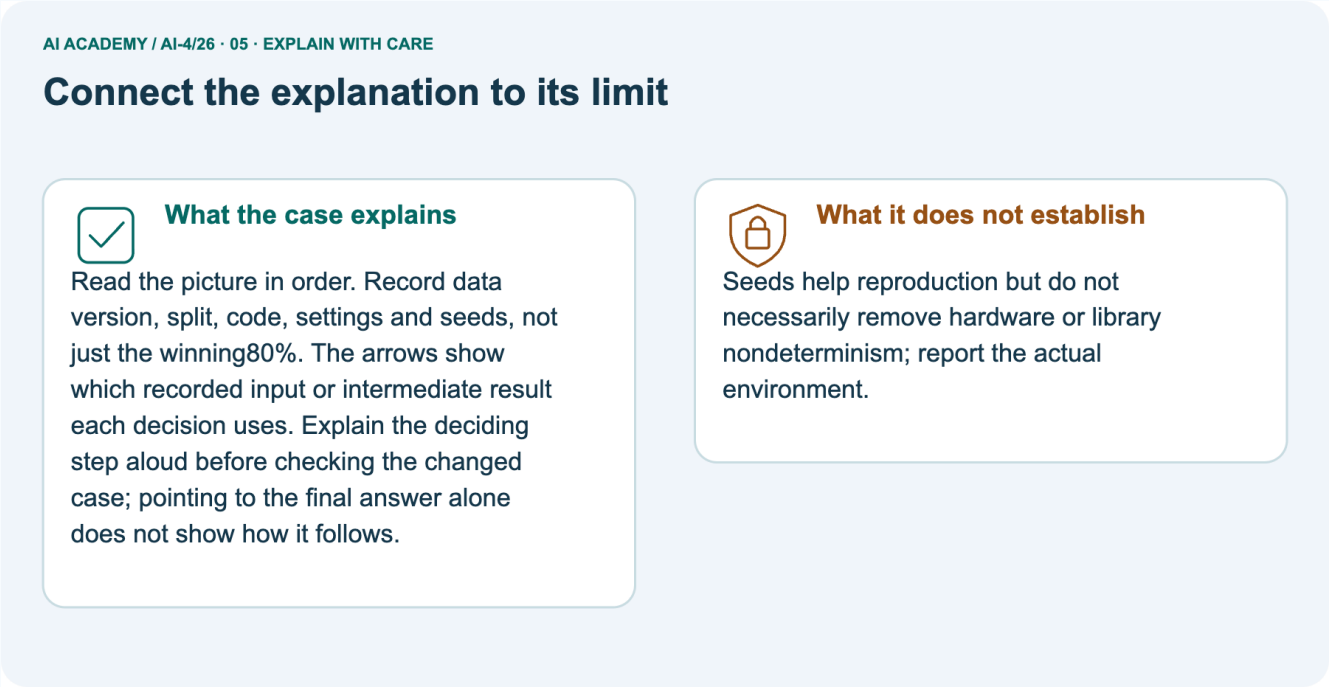
Original case: Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. Changed condition: The report deletes the **60%** run.

Expected result: The remaining mean rises to75%, but the reported comparison becomes selectively incomplete. Reason: Removing an unfavourable declared run changes the **evidence** rather than improving the method.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

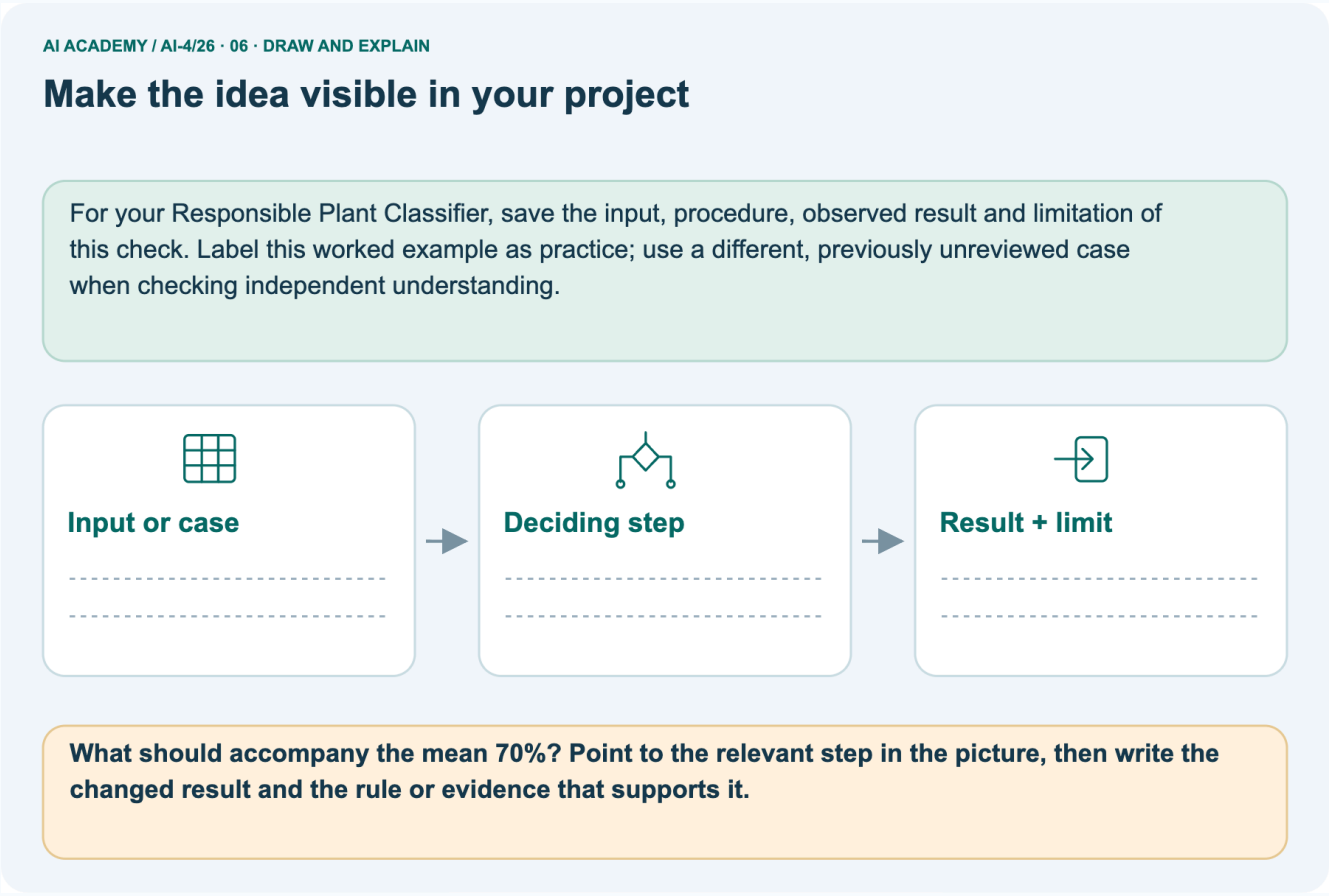
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: **Seed**s help reproduction but do not necessarily remove hardware or library nondeterminism; report the actual environment.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should accompany the mean **70%**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Reproducible Experiments: the case

02. Follow the deciding steps

03. All declared seeds

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.